Written Preliminary Examination

The Ph.D. Written Preliminary Examination is a research proposal prepared by each Ph.D. student that provides suitable motivation for the proposed doctoral research, a summary of accomplishments, and a detailed research plan of thesis research.

Timing

For students starting Fall 2021 or later, the Ph.D. Written Preliminary Exam will be taken in the fifth semester of a student's Ph.D. studies. Failure to submit the exam on time is grounds for dismissal from the program unless the advisor provides a written motivated request to the DGS for a one semester (and only one semester) delay. Exceptions can be made by the DGS for extenuating circumstances. The student may elect to take the Written Preliminary Exam earlier than their fifth semester if they have completed all ME course requirements. Students may register for Ph.D. pre-thesis credits (up to the 14 credit allotment) during the semester they take their Written Preliminary Exam. The Written Preliminary Exam must be submitted to the Graduate Program Manager, along with their advisor and committee members on October 15 for the Fall semester or March 15 for the spring semester. After submission, the Graduate Program Manager will check it to ensure the exam meets the formatting requirements.

Examination Committee

The examination committee for the Written Preliminary Examination consists of the student's advisor(s) and at least two additional members of the Mechanical Engineering Graduate Faculty (members or senior members). Selection of the examination committee is at the discretion of the student with approval of the advisor(s). Members of the Written Preliminary Examination committee will also serve on the oral examination committee.

Evaluation

A three-week reading period is required for the committee to evaluate the Written Preliminary Examination and provide feedback. The Written Preliminary Examination will be evaluated on the basis of the scientific and technical comprehension, the quality of the research plan, and the ability to communicate clearly and effectively. Faculty examiners will provide feedback using the google form, sent with the WPE. The feedback form includes the option for examiners to pose up to five written questions for the student to address during the Oral Preliminary Examination. The examiners, including the advisor, will submit their forms to the Graduate Program Manager. Once all the forms are submitted, the Graduate Program Manager will submit the evaluation form package to the student with a copy to the Oral Preliminary Examination Committee (consisting of the written exam committee and one non-ME faculty member).

The examiners will indicate on the evaluation form their assessment of the exam using scoring options of pass, revision or fail. If one or more members of the committee score "revision" the student needs to resubmit the written exam with the requested revisions within 2 weeks after receiving the feedback (please submit in addition to a clean pdf document also a document with track changes enabled). If one or more members of the committee score "fail," the Written Preliminary Exam must be revised with a due date of the following semester, leading to a one semester delay in the oral examination. The need to revise the exam will not interfere with the student's ability to enroll in thesis credits. When assessing the revised written preliminary examination, a "pass" score by two of the three members of the examination committee constitutes a passing grade (if there are four committee members, three of four must approve the revision). Students who do not receive a passing grade after revision will be terminated from the Ph.D. program.

Format

Students are expected to write the thesis proposal independently in consultation with the faculty advisor(s). The advisor is encouraged to work directly with their student on defining and formalizing the objectives of their proposed research and on defining and describing the methods to be utilized in the proposed research. The advisor is welcome to assist the student in editing the written preliminary examination grammatically and structurally, but the bulk of the writing needs to come from the student. Plagiarism on the written examination will result in immediate dismissal from the program, including plagiarism of portions of prior written examinations. Any figures taken from peer-reviewed manuscripts, prior examinations, or proposals need to be fully cited.

The proposal must be legible, and conform to the following requirements:

- The font size should be easy to read. Please use Times New Roman or a Computer Modern Family Font at 11 point font or larger, Arial, Courier New, Calibri, or Palatino Linotype at a font size of 10 point or larger.
- A font size of 10 point or less may be used for mathematical formulas or equations, figure, table or diagram captions and when using a Symbol font to insert Greek letters or special characters. Please note, however, that the text must still be readable at 100% size. Figure captions can have a different font style, such as Arial, to help offset them from the main text.
- All margins must be at least an inch.
- Page limits and line spacing requirements apply. See below.

Adherence to these formatting guidelines will be reviewed by the Graduate Program Manager. Proposals not meeting the formatting guidelines will be returned and the examination will be delayed a semester.

Content

The proposal must contain the following sections:

- ***Cover Page***. The Cover Page must contain the title of the proposal, student name, advisor(s)'s name, and the names of the student's committee members, including which ones will serve as readers. The submission date should also be included.
- ***Project Summary***. The Project Summary is a one page, single-spaced summary of the proposed activity. The Project Summary should be a self-contained description of the proposed research. It should include a statement of objectives and methods to be employed. It must address the scientific and/or technical merit of the project, for example, the influence that the results might have on the direction, progress and thinking in relevant scientific or engineering fields. It must also address the appropriateness of the proposed methods, and the logic and feasibility of the research approach. The Project Summary should be informative to persons working in the same or related fields and, insofar as possible, understandable to a scientifically or technically literate lay reader.
- ***Project Description***. The recommended length for the double-spaced Project Description is 20-25 pages (30 page maximum). It should include ***background & motivation, objectives***, ***proposed research***, ***preliminary results***, and ***conclusion***. Multiple proposed research and preliminary results sections may be used if the proposed research is best divided up into sub-topics. The precise organization of these sections is up to the student, in consultation with the advisor(s). Figures and tables (with captions) are included in the 30 page limit.
    - The ***background & motivation*** section should include a critical review of the relevant literature. It should indicate the current state of understanding of the proposed research topic. The review should

emphasize how the prior work relates to the proposed study. It should address the significance and limitations of the cited work. If any figures are taken from another publication or a website, they must be referenced explicitly in the caption.

- The ***objectives*** section should provide a context for the proposed work as well as specific objectives and expected significance of the Ph.D. dissertation. It should contain a brief overview of the proposed work in relation to the present state of knowledge in the field and to work in progress by others. This section should be two pages or less.
- The ***proposed research*** section (or sections) is the key element of the proposal and should comprise no less than half the page limit set for the project description. This section should outline the plan of work, including the activities to be undertaken, and a description of experimental and/or computational methods and procedures. It should address what will be done, how it will achieve the proposed objectives, and should contain justification for the approach proposed. The proposed research section should complete the arguments developed earlier and present initial ideas on how to solve the problems posed. Strong proposals avoid repetition and digression. This section (or sections) should relate the proposed activities to the project objectives and provide expected outcomes, including how the proposed research, if successful, will contribute to a greater understanding of the topical area. An assessment of risk should be provided with the proposed research as well as a contingency plan in case a particular research avenue proves inexpedient in some manner. It should present a reasoned path from where the student begins to where they want to be at the end of the research.
- The ***preliminary results*** section (or sections) should provide an overview of completed work including its significance. Note that preliminary results are not required.
- The ***conclusion*** section should provide a concise summary of what will be known when the research has been completed that is not known. It must present the scientific and/or technical merit of the project.

- ***References.*** Reference information is required. Each reference must include the names of all authors (in the same sequence in which they appear in the publication), the article and journal title, book title, volume number, page numbers, and year of publication. Students must be especially careful to follow accepted scholarly practices in providing citations for source materials, including websites, relied upon when preparing any section of the research proposal. There is no page limitation for the references, but this section must include cited references only. References should be single-spaced, but a blank line between each individual reference is preferred.
- ***Biosketch***. A biosketch of not more than 1 page, single-spaced, should be provided. The biosketch should be written in a narrative format and it should include a summary of previous education, including institutions attended, and research experiences.
- A copy of the approved Graduate Degree Plan.